Hans Aaron Laureles

AI SYSTEMS ENGINEER & FULL-STACK BUILDER

Cavite, Philippines | hanslaureles92@gmail.com | hansaaronlaureles.netlify.app | BS Computer Science (2021–2025)

EXECUTIVE PROFILE

AI Systems Engineer and BS Computer Science graduate (DLSU-D 2021–2025) specializing in autonomous multi-agent swarms, zero-dependency cognitive memory engines, local GPU inference pipelines (AMD DirectML), and production full-stack systems with executive design craft.

PROFESSIONAL & SYSTEMS EXPERIENCE

Lead AI Systems Architect · LSFM AI HQ — 2025 – Present · Autonomous Workstation

Architected an autonomous 5-agent platform over Discord Gateway with hybrid inference routing (Groq Qwen 2.5 70B + local AMD RX 6600 XT via DirectML) and programmatic OAuth2/headless tools at $0 cloud cost.

Lead Systems Architect · Capstone: Patriot — 2023 – 2024 · DLSU–D

Engineered real-time matchmaking mobile app in React Native (TypeScript) with AWS Lambda, DynamoDB, Redis caching for sub-50ms queries, and custom algorithmic relevance scoring.

UI/UX & Frontend Engineer Intern · ROC (roc.ph) — June 2024 – August 2024 · Cavite, PH

Architected responsive design token frameworks in Figma and shipped accessible, semantic HTML5, CSS3, and React components across production client applications.

ROLE-TAILORED ACHIEVEMENTS & SYSTEMS IMPACT

- Architected zero-dependency cognitive memory core executing sub-15ms BM25 heuristic recall before code generation, eliminating AI session amnesia with 100% rule retention.
- Orchestrated 5 concurrent autonomous daemons over Discord Gateway with hybrid local/cloud LLM failover at $0.00/month cloud compute cost.
- Engineered headless Chromium compilation pipeline generating strict single-page ATS-verified vector PDFs in under 900ms.
- Designed high-conversion specialty e-commerce flow with client-side drink configurator modal, sliding bag drawer, and Philippine payment rails (GCash, Maya, QR Ph).

FLAGSHIP PORTFOLIO CASE STUDIES

LSFM AI HQ — Autonomous Multi-Agent Swarm & Hybrid Inference

Orchestrated 5 Discord agent daemons with hybrid local AMD DirectML / Groq Cloud failover, headless PDF compiler, and OAuth2 triage.

Cognitive Memory Core — Zero-Dependency Self-Improving Agent Flywheel

Engineered sub-15ms BM25 episodic memory recall across past failure symptoms, post-mortem reflection logging, and rule crystallization.

Aura Coffee & Kitchen — Specialty Commerce & Philippine Payment Rails

Built client-side drink configurator modal, real-time sliding bag drawer, and 1-page checkout tailored for GCash, Maya, and QR Ph.

EDUCATION

Bachelor of Science in Computer Science (BS CS) — 2021 – 2025

De La Salle University–Dasmariñas (DLSU-D) — Distributed Systems · Human-Computer Interaction · Algorithms

TECHNICAL PROFICIENCIES & CAPABILITIES

Applied AI & Agents:	Multi-Agent Swarms, Cognitive Architecture (BM25 Recall), Local GPU Inference (DirectML/Vulkan), Ollama, Groq Cloud, Tool Calling, Structured JSON
Systems & Automation:	Python 3.11 (asyncio, stdlib), Google Workspace REST APIs, Headless Chromium, AWS (Lambda, DynamoDB), Redis Caching, Git CI/CD
Interface Engineering:	React, Next.js, TypeScript, React Native, Design Tokens & CSS Vars, Tailwind CSS, Local Payments (GCash, Maya, QR Ph), Figma